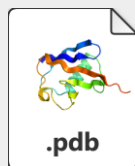




User

Prompt: Starting from a seed SMILES and a receptor pocket, propose improved analogs under basic constraints (potency up, lipophilicity not too high) and return a ranked shortlist.



1RX2.pdb

COc1ccc(cc1)C(=O)NCCN2CCOCC2
Seed SMILES

Output

Final Ranked Shortlist of Improved Analogs



#1: COc1ccc(C(=O)NCCc2nn(C(F)(F)F)cc2)cc1
#2: COc1ccc(C(=O)NCCN2CCc3ccccc3C2)cc1
#3: COc1ccc(C(=O)NCCc2noc3c2CCCC3)cc1
.....



Orchestrate Agent



Decompose & Plan
Plan Task

Delegate:
Target Analysis



Retrieve Agent

Receive Task1
Analyze 1RX2.pdb

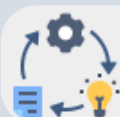


Tool Call: PDB Analysis
Identify pocket residues



Target Info:
DHFR, Pocket Residues

Report: Target Info



Track State & Refine

Delegate:
Generate docked conformations



Forge Agent

Receive Task2
SAR & Design



Tool Call: PocketFinder
(Identify Worst Fragment)



Tool Call: FragGen
(Generate 5 Analogs)

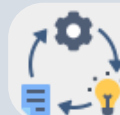


Tool Call: PLIP
(Interaction Analysis)



Ranked Shortlist:
5 Analogs with predicted interactions

Report: Analogs & Interactions



Track State & Refine



Synthesize Results & Formulate Response



Gauge Agent

Receive Task3
Quantitative Evaluation



Tool Call: ADMET-AI
(Calculate LogP, QED, etc.)



Evaluation Data:
ADMET profiles (LogP, QED...)

Report: Evaluation Results